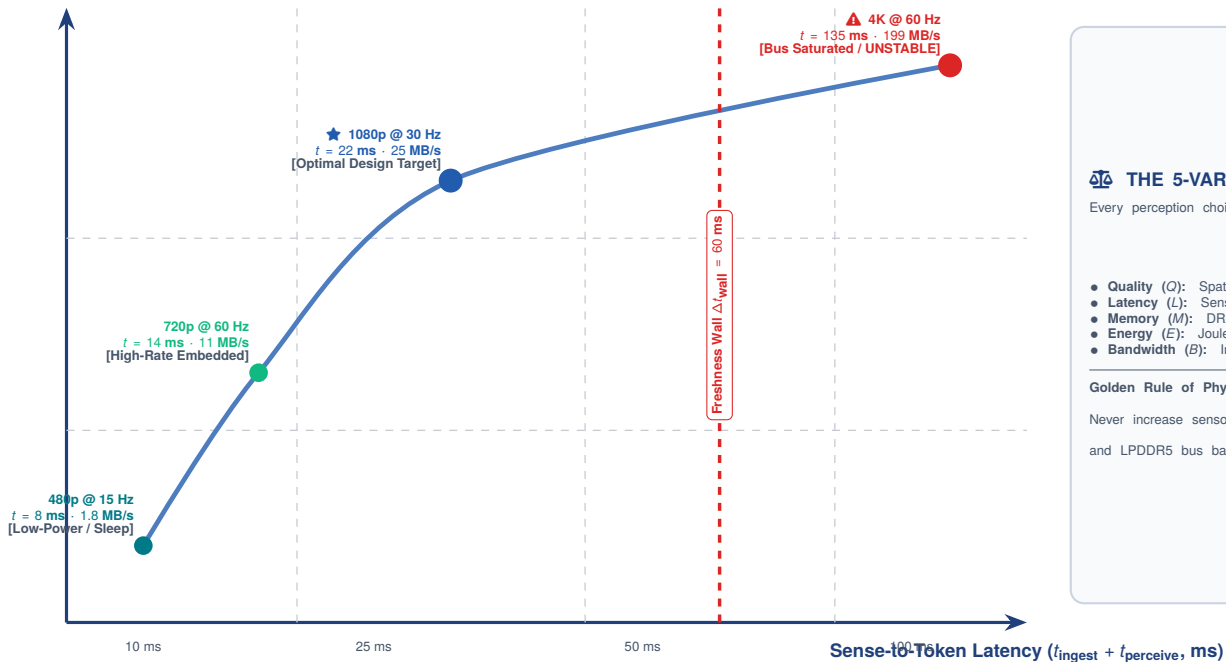


THE MULTI-OBJECTIVE PERCEPTION OPERATING FRONTIER

Navigating the Pareto Trade-Off: Spatial Precision vs. Ingestion Latency, Memory Bandwidth, and Bus Saturation

Spatial Quality / Metric Precision (Q)



THE 5-VARIABLE PARETO TRADE-OFF

Every perception choice balances:

$$P_{\text{Opt}} = f(Q, L, M, E, B)$$

- **Quality (Q):** Spatial patch resolution for millimeter grasp precision.
- **Latency (L):** Sense-to-token duration entering total loop delay t_{total} .
- **Memory (M):** DRAM ring buffer footprint vs. SRAM NPU tiling.
- **Energy (E):** Joules per token extraction (thermal envelope ≤ 15 W).
- **Bandwidth (B):** Ingestion traffic vs. shared UMA crossbar limit.

Golden Rule of Physical AI:

Never increase sensor resolution Q unless the latency budget $\Delta t \leq \tau_{\text{world}}$ and LPDDR5 bus bandwidth can sustain it without P_{99} jitter.